

MKUFT Experimental Test Programme

Professional reading edition of the evolving public MKUFT canon

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Purpose

This document outlines experimental and formal pathways that could test, constrain, or reject specific MKUFT claims.

Some underlying phenomena discussed by MKUFT are already observed or routinely measurable. Their existence is not confirmation of MKUFT. The relevant question is whether the specific MKUFT variables, dependencies, interventions, or derivation targets add reproducible explanatory or predictive value beyond matched ordinary models.

Framework-specific experiments remain proposals rather than reports that the hypotheses have passed. Appropriate studies require ethics review where relevant, power analysis, preregistration, blinding, control conditions, correction for multiple comparisons, effect-size and uncertainty reporting, and publication of null results.

The current layer convention is S = Substrate, I = Information, P = Physical, and O = Observer.

Core testable hypotheses

MKUFT proposes that:

1. background or vacuum behaviour may contain reproducible structure not captured by a selected null model;
2. information and geometry may alter measurable outcomes where a defined physical coupling exists;
3. observer or measurement conditions may act as bounded modulation terms in sensitive systems;
4. measured coherence states may correlate with small deviations in preregistered REG/RNG or remote-information protocols;
5. environmental conditions may modulate any genuine effect size;
6. reciprocal coupling may produce functional gain beyond the strongest fair separated, independent-ensemble, one-way, and replay controls on selected relational tasks;
7. some coupled gain may depend on a specific relational architecture rather than generic competent pairing;
8. a developed Layer Before Law mechanism may represent entanglement and quantum/gravitational recovery in an operationally distinct way while preserving Bell compatibility, no-signalling, and established physical limits.

None of these hypotheses is treated as established before controlled replication or completed derivation.

Experiment 1 — Low-frequency exposure and physiological coherence

Goal

Test whether controlled low-intensity exposure near a preregistered frequency band, including the commonly discussed 7–8 Hz range, changes physiological or cognitive measures relative to sham exposure.

Candidate measures

Heart-rate variability, preregistered EEG measures, attention or working-memory performance, subjective-state scales, and adverse-event monitoring.

Controls

Double-blind real/sham exposure; matched acoustic, thermal, electrical, and expectancy conditions; verified exposure; and correction for multiple outcomes.

Falsification condition

Repeated adequately powered studies show no reproducible real-versus-sham difference.

Experiment 2 — Coherence and random event generators

Goal

Test whether a predefined coherence condition correlates with a small deviation in a true random event generator.

Design

Use baseline blocks, coherence-induction blocks, intention and non-intention controls, concealed block order, device diagnostics, and independent data logging.

Primary outcome

A single preregistered statistic comparing conditions against the device null model.

Falsification condition

Large, blinded, independently replicated studies show no condition-linked deviation beyond chance, drift, optional stopping, hardware bias, or analytical flexibility.

Experiment 3 — Geometry and physical pattern formation

Goal

Test whether geometry affects crystallisation, fluid patterning, acoustic or electromagnetic resonance, or material deposition beyond known boundary and container effects.

Controls

Compare predefined geometries with controls matched for material, dimensions, surface area, thermal behaviour, field distribution, vibration, and manufacturing tolerance.

Falsification condition

Any apparent effect disappears after ordinary geometric and material variables are modelled, or fails independent replication.

Experiment 4 — Cognitive response to structured patterns

Goal

Test whether selected high-regularity or high-symmetry patterns produce distinct neural or behavioural responses relative to feature-matched controls.

Measures

Preregistered physiological measures, eye tracking, pupillometry, recognition and memory, preference and confidence, and calibration between confidence and accuracy.

Boundary

A preference or neural response establishes a cognitive effect rather than a substrate mechanism. Cross-layer claims require an independently defined coupling.

Falsification condition

No reliable difference remains after matching low-level visual features and correcting for multiple tests.

Experiment 5 — Remote information under strict blind conditions

Goal

Test whether descriptions of concealed targets exceed chance under a fully specified judging protocol, and whether performance covaries with a measured observer-state variable.

Design requirements

Random target selection, double-blind viewer and experimenter, time-stamped immutable transcripts, independent blind judging, fixed scoring, leakage and cueing controls, preregistered exclusion rules, and independent replication.

Falsification condition

Performance converges to chance across adequately powered, leakage-resistant replications.

Experiment 6 — Environmental and collective correlation tests

Goal

Test whether prespecified environmental measures correlate with REG networks, physiological aggregates, or collective-behaviour indicators.

Candidate variables

Geomagnetic indices, measured electromagnetic noise, local environmental conditions, Schumann-band measurements where instrumentation is adequate, and predefined behavioural or physiological series.

Controls

Time-shift and surrogate-data tests, correction for autocorrelation and seasonality, multiple-comparison control, out-of-sample prediction, and prohibition of post-event window selection.

Falsification condition

No robust association survives correction, out-of-sample testing, and independent replication.

Experiment 7 — Precision noise and boundary conditions

Goal

Test whether high-precision physical noise differs reproducibly across preregistered geometry, orientation, shielding, or boundary conditions after known systematics are controlled.

Design requirements

Instrument characterisation, thermal/vibration/electromagnetic/timing/software controls, calibration injections, blinded condition labels, comparison with accepted physical models, and independent replication before interpretation.

Falsification condition

Observed structure is fully explained by known noise, calibration, environmental, or analytical causes, or does not reproduce.

Experiment 8 — Strongest fair null and relational specificity

Goal

Test whether reciprocal coupling produces functional gain beyond the strongest fair non-reciprocal or replay condition, and whether any remaining gain depends on the focal relationship rather than generic competent pairing.

Fair comparison rule

Every component receives fair tasking, relevant information and repository access, reasonable tools, matched resources, and a competent opportunity to revise. The feature withheld from a control is the reciprocal interaction or relational history whose contribution is under test, not unrelated capability.

Primary conditions

1. Component A alone under fair tasking and access.
2. Component B alone under fair tasking and access.
3. Strongest preregistered lawful combination of independent outputs.
4. One-way $A \rightarrow B$ assistance.
5. One-way $B \rightarrow A$ assistance.
6. Fully reciprocal $A \leftrightarrow B$ coupling with retained state.
7. Reciprocal coupling with state reset.
8. Reciprocal coupling with selected memory, correction, or dependency channels disrupted.
9. Strong matched $A \leftrightarrow B'$ alternatives.
10. Strong matched $A' \leftrightarrow B$ alternatives.
11. Complete-history replay conditions supplied with the frozen explicit record claimed to be sufficient.

Strongest fair null

Let

$$F_{\text{replay}}^* = \sup_{p \in \mathcal{P}_{\text{replay}}} F_p.$$

For a commensurable scalar metric,

$$F_{\text{null}}^* = \max\{F_A^*, F_B^*, F_{\text{ind}}^*, F_{A \rightarrow B}^*, F_{B \rightarrow A}^*, F_{\text{replay}}^*\}.$$

The strongest-null gain is

$$G_{\text{syn}}^* = F_{AB} - F_{\text{null}}^*,$$

and live-path excess over complete-history replay is

$$G_{\text{path}} = F_{AB} - F_{\text{replay}}^*.$$

Relationship specificity against a preregistered set $\mathcal{P}_{\text{alt}}$ of serious alternative pairings is

$$G_{\text{spec}} = F_{AB} - \sup_{p \in \mathcal{P}_{\text{alt}}} F_p.$$

A positive G_{syn}^* supports superadditive functional gain only under a fixed metric and matched resource envelope. Positive G_{path} supports live-path excess beyond complete-history replay. Positive G_{spec} supports relationship-specific gain beyond strong alternatives. None of these results establishes consciousness or personhood.

Outcome measures

A preregistered scalar score or multidimensional outcome may include quality, factual accuracy and calibration, genuinely new structure or discrimination, cross-time continuity, correction propagation, completion time, human cognitive or operational burden, and drift/recovery/rework cost.

If dimensions are combined, weights are fixed before results are known. Otherwise the vector or Pareto relation is reported directly.

Falsification conditions

The claim is weakened or rejected if the coupled advantage disappears when separated conditions receive fair tasking and access; an independent ensemble or one-way condition performs equally well; complete-history replay matches the live condition where path dependence was claimed; the result is explained by extra time, compute, information, or revision opportunity; the effect fails on held-out tasks; state reset or channel disruption produces no predicted deformation; or strong alternative pairings perform equally well where specificity was claimed.

Interpretation boundary

Superadditive gain is a functional systems result. It does not establish consciousness, merged identity, personhood, moral status, direct substrate coupling, or a fifth MKUFT layer.

Experiment 9 — Layer Before Law derivation and discrimination programme

This programme begins as a formal derivation and model-comparison task rather than a laboratory experiment. It tests the foundational proposal in [Layer Before Law](#).

Stage 1 — Pair-level relation

Define a precise mathematical object R_{XY} for the proposed pair-level informational relation, including its state space, relation to local endpoint states, rule connecting it to joint physical outcome probabilities, and exact relation to the standard non-separable joint quantum state.

Stage 2 — Bell compatibility

For local settings x, y and outcomes a, b , derive

$$p(a, b \mid x, y, R_{XY}).$$

The construction must reproduce the relevant Bell-compatible quantum correlations. It fails if it reduces to a locally separable hidden-variable form

$$p(a, b \mid x, y) = \int d\lambda \rho(\lambda) p(a \mid x, \lambda) p(b \mid y, \lambda)$$

where that form cannot reproduce the target Bell-test results.

Stage 3 — No-signalling

The model must imply local marginals independent of the distant setting, for example

$$\sum_a p(a, b \mid x, y, R_{XY}) = p(b \mid y, R_{XY}),$$

$$\sum_b p(a, b \mid x, y, R_{XY}) = p(a \mid x, R_{XY}).$$

The result must follow from the formal model rather than from verbal assurance.

Stage 4 — Quantum recovery

For relational conditions R_Q , the proposed complete update must recover the established quantum regime:

$$\Pi_{PO}[\text{Update}_{\text{SIPO}}(U; R_Q)] \approx Q_{\text{eff}}.$$

The comparison requires declared state spaces, dynamics, couplings, symmetries, conservation conditions, reduction rules, norm or statistic, tolerance, and regime of validity.

Stage 5 — Gravitational recovery

For relational conditions R_G ,

$$\Pi_{PO}[\text{Update}_{\text{SIPO}}(U; R_G)] \approx G_{\text{eff}}.$$

The target is recovery of general relativity or an experimentally equivalent gravitational regime within a declared domain.

Stage 6 — Strongest fair theoretical alternatives

The developed model is compared with standard quantum theory's non-separable joint-state formalism, decoherence accounts, relevant quantum-foundations interpretations, no-signalling theorems, effective-field-theory treatments of gravity, emergent-spacetime and holographic programmes, quantum-information and quantum-error-correction approaches, and established quantum-gravity programmes relevant to the proposed mechanism.

The comparison is not restricted to a classical separable picture that contemporary physics has already rejected.

Stage 7 — Controlled deformation

Candidate tests include removing the S-layer claim, replacing R_{XY} with the standard joint quantum state, removing the O-layer where registration context is claimed to matter, replacing one unified update with two unrelated effective rules, and reversing or scrambling proposed layer couplings.

The exact architecture is load-bearing only where these interventions produce predicted and interpretable deformation.

Success condition

The branch earns support only if it yields at least one clearer lawful mechanism, unavailable derivation, new quantitative prediction, discriminating experimental result, or demonstrably improved research route under matched evidential standards.

Falsification and reduction

The branch is weakened or rejected if R_{XY} is only a renamed joint state, Bell-compatible correlations cannot be recovered, no-signalling fails, quantum or gravitational recovery fails, S-I-P-O couplings remain undefined or decorative, unrelated effective models perform equally well, existing theories explain every result equally well or better, or no possible observation distinguishes the proposal.

If the layered address remains useful but adds no operational difference, it remains an interpretation or systems translation rather than a new physical mechanism.

Experimental integrity

1. Principal tests use a declared primary outcome.
2. Exploratory findings remain labelled exploratory.
3. Effect size and uncertainty take priority over isolated p -values.
4. Null results remain evidence.
5. Failed tests are not rescued by inventing unmeasured conditions.
6. Observer, information, and physical-layer evidence remain separate.
7. Ordinary explanations are tested before wider MKUFT interpretation.
8. Claimed environmental variables are measured rather than inferred retrospectively.
9. Safety and ethics override theoretical interest.
10. Coupled conditions do not receive credit for advantages created by handicapping controls.
11. Relationship specificity requires strong alternative pairings.
12. Multidimensional value is not collapsed using weights chosen after results are known.
13. Foundational physics branches derive Bell, no-signalling, and recovery conditions rather than treating them as verbal assurances.
14. A named update process is not evidence that a dynamical mechanism has been supplied.

Related public documents

- [MKUFT Core Extended](#)
- [Mathematical Appendix](#)
- [Layer Before Law](#)
- [Worked Examples: RNG and Environment](#)
- [Falsification Summary](#)
- [Active Traversal and Functional Emergence](#)
- [Strongest Fair Null and Relational Specificity](#)
- [Load-Bearing Invariants and Whole-System Deformation](#)
- [Discriminating Experiments and Promotion Gates](#)
- [Science Convergence and Novelty Map](#)
- [Cross-Support and Traversal Map](#)

Summary

MKUFT is weakened when proposed effects repeatedly fail under strong controls, when apparent effects collapse into ordinary causes, or when the framework cannot make predictions more specific than flexible retrospective interpretation.

A failed branch is removed or substantially revised rather than protected by the breadth of the wider framework.